FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

IT0049
(WEB SYSTEM TECHNOLOGIES)

TECHNICAL FORMATIVE ASSESSMENT

6

FROM ARRAYS TO ELOQUENT

Student Name / Group Name:	Tristan Acha
Members (if Group):	**Name** / **Role**
Section:	TC37
Professor:	Mr. Ric Cristian Frias

PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements. [PO: C]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Apply advanced web development principles in creating, debugging, validating, and securing database-backed web applications. [CLO: 2]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Create Laravel migrations for the customers and users tables.
- Create and run a seeder to populate sample data.
- Build Eloquent models for both tables.
- Retrieve and display records in a Blade view.

BACKGROUND INFORMATION

This activity replicates TFA2 in Laravel. Instead of hand-writing SQL to create tables, Laravel migrations describe the schema as PHP code that can be run, rolled back, and shared through version control. A seeder then populates sample data automatically, and Eloquent — Laravel's ORM — retrieves records as PHP objects without raw queries.

GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (40 points max)	All required pages and features are present and work as specified in the activity instructions.
Code Structure & Organization (25 points max)	Controllers, models, and views are separated correctly and follow the framework's conventions; code is readable and sensibly named.
Database Design & Query Usage (20 points max)	Migrations correctly create the specified schema, the seeder populates realistic sample data, and records are retrieved through Eloquent models and displayed correctly in the view.

CRITERIA	Description
Documentation & Submission Quality (15 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.

LABORATORY ACTIVITY

Rebuild TFA5's POS application with a real database: replace the static arrays with migrations, a seeder, and Eloquent models, using the same schema as TFA2.

1. Create migrations for the customers table (full_name, email, phone) and the users table (username unique, full_name), matching TFA2's schema.
2. Create a seeder that populates at least 5 sample records in each table.
3. Create Customer and User Eloquent models.
4. Update your CustomerController and UserController to retrieve records through their Eloquent models instead of static arrays.
5. Confirm the Blade views display the retrieved records the same way they displayed the static array data in TFA5.

Submission Requirements:

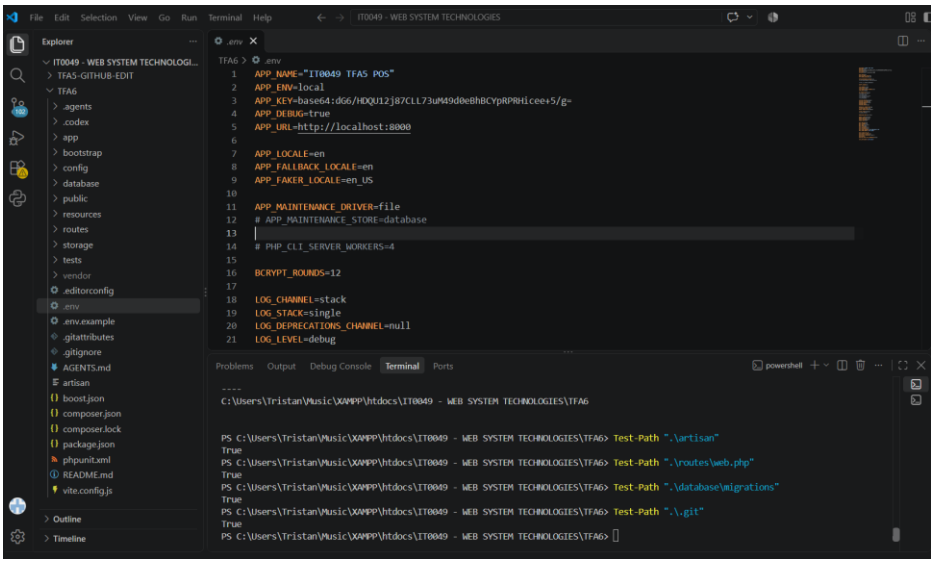
- A link to a GitHub repository containing your raw project files, including your database export or migration and seeder files.
- A link to your hosted, working version of the application.

REFERENCES

- Stauffer, M. (2024). Laravel: Up & Running: A Framework for Building Modern PHP Apps (3rd ed.).
- Laravel Documentation. Migrations. https://laravel.com/docs/13.x/migrations
- Laravel Documentation. Eloquent — Getting Started. https://laravel.com/docs/13.x/eloquent
- Laravel Documentation. Seeding. https://laravel.com/docs/13.x/seeding

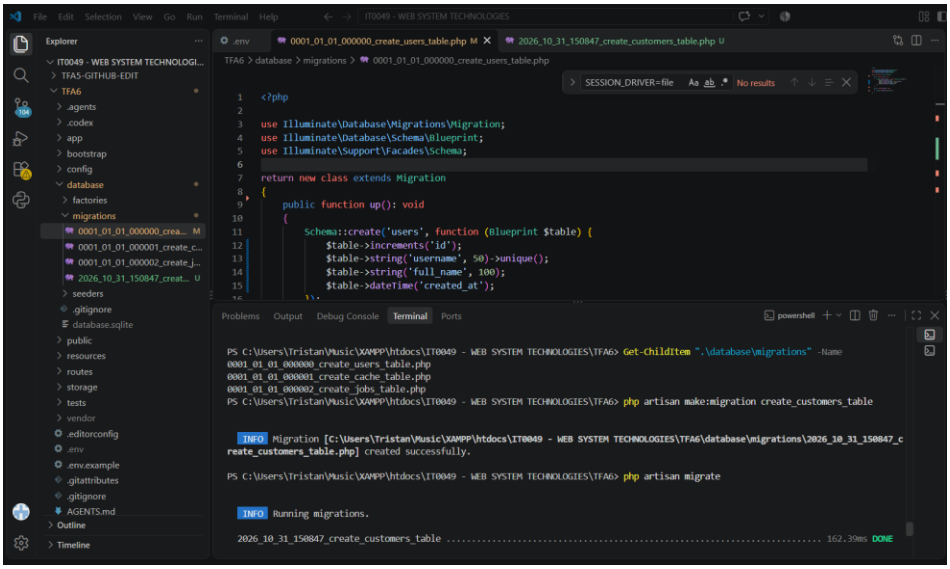
Project Preparation

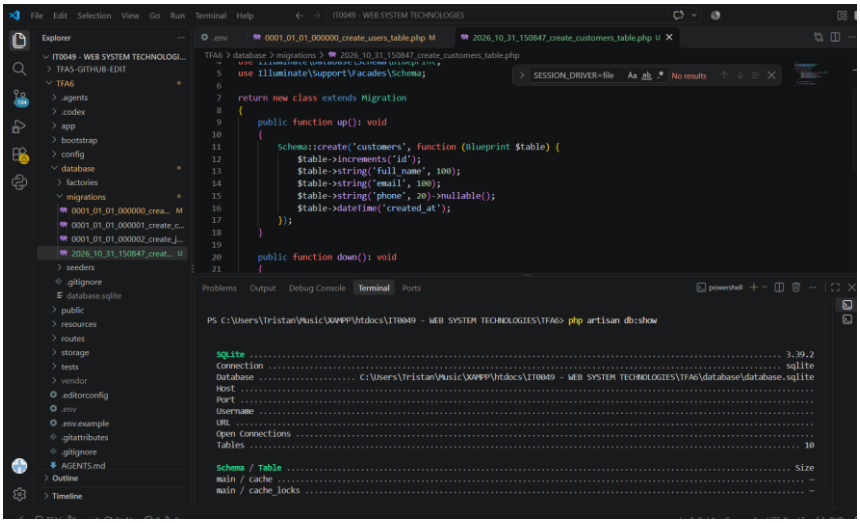
The existing TFA5 Laravel POS application was copied into a separate TFA6 project directory to preserve the previous activity. This project will replace the temporary PHP arrays with database migrations, seeders, and Eloquent models based on the TFA2 database schema.

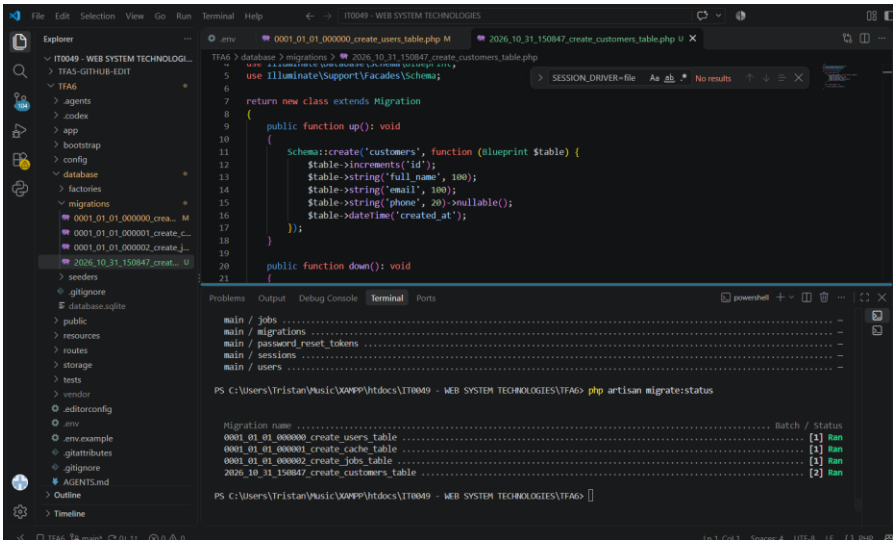


1. Database Migrations

Created and executed Laravel migrations for the Customers and Users tables using the TFA2 database schema. The tables include the required fields, primary keys, unique username constraint, and nullable phone number. The SQLite database was verified using Artisan commands to confirm that both tables were created with the correct structure.

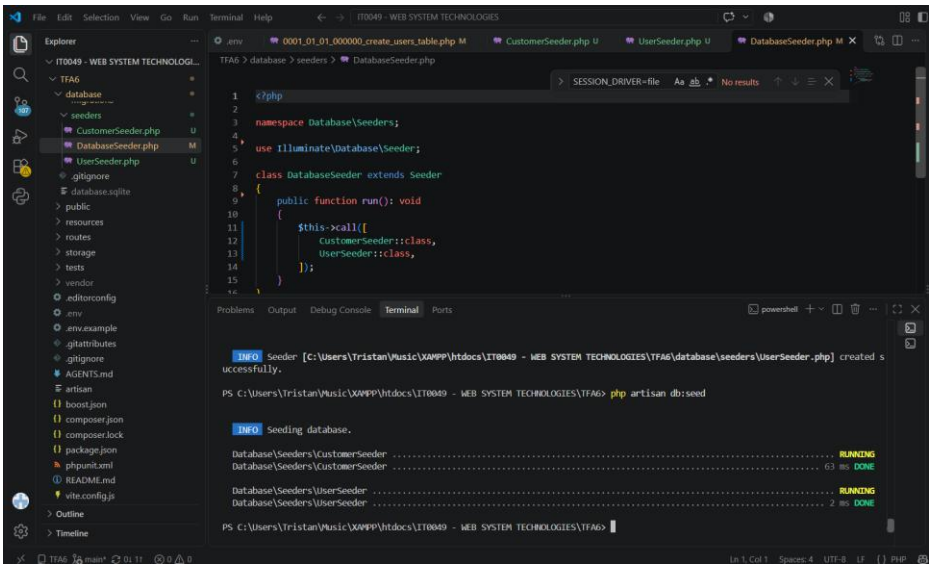


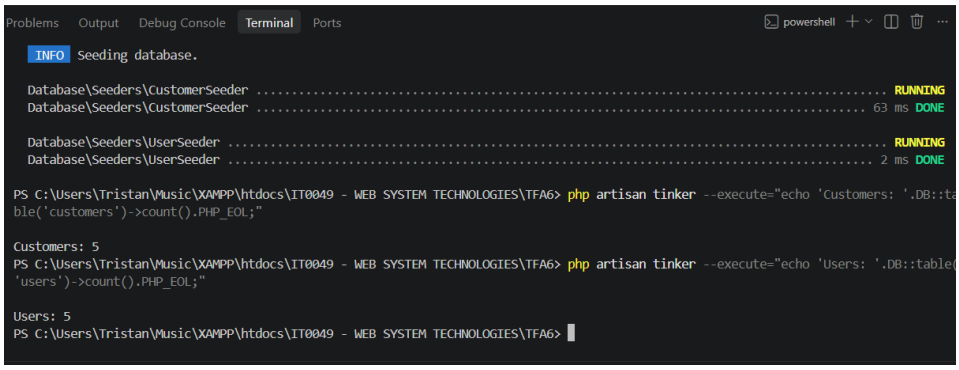




2. Database Seeding

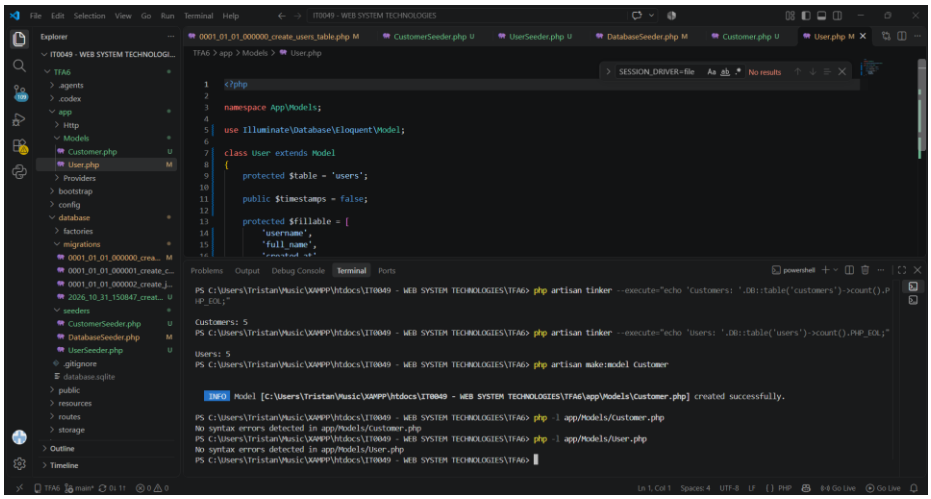
Created CustomerSeeder and UserSeeder using five sample records for each table based on the previous POS activity. Both seeders were registered in DatabaseSeeder and executed successfully. Laravel Tinker confirmed that the database contains five customer records and five user records.

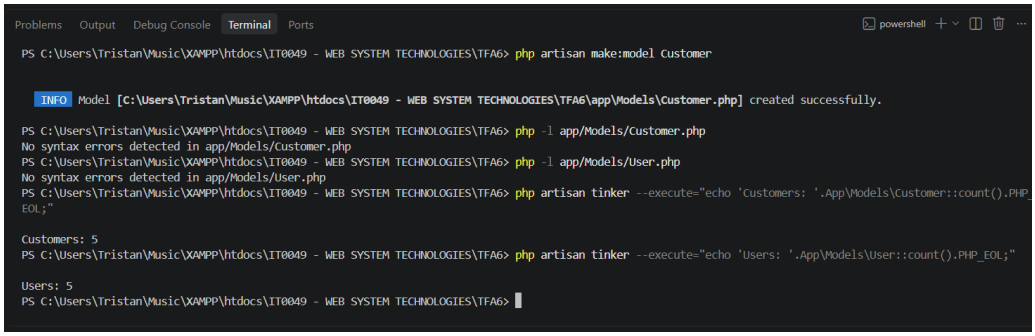




3. Eloquent Models

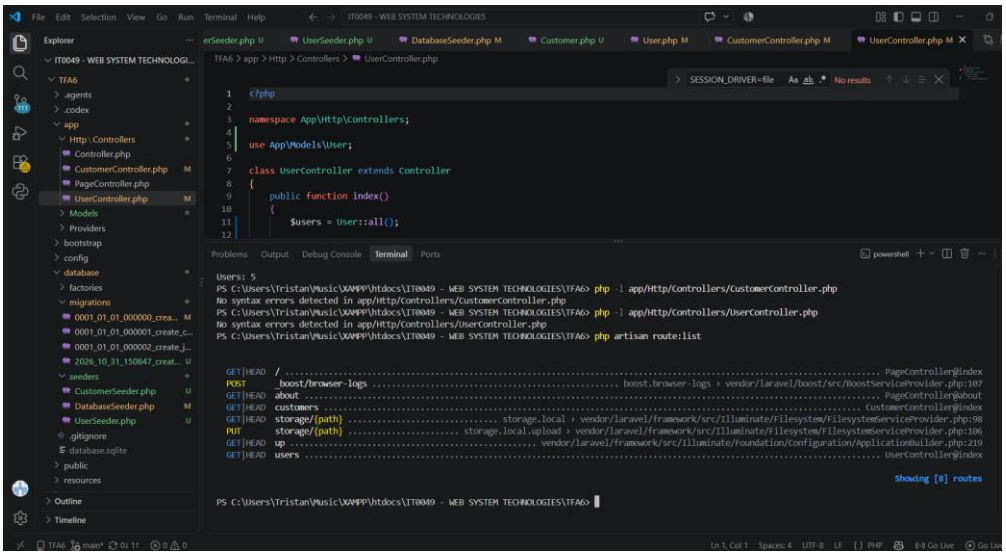
Created the Customer model and updated the User model to match the TFA2 database schema. Both models were configured to retrieve database records using Eloquent ORM. Syntax checks and Eloquent queries confirmed that both models work correctly and return five records each.





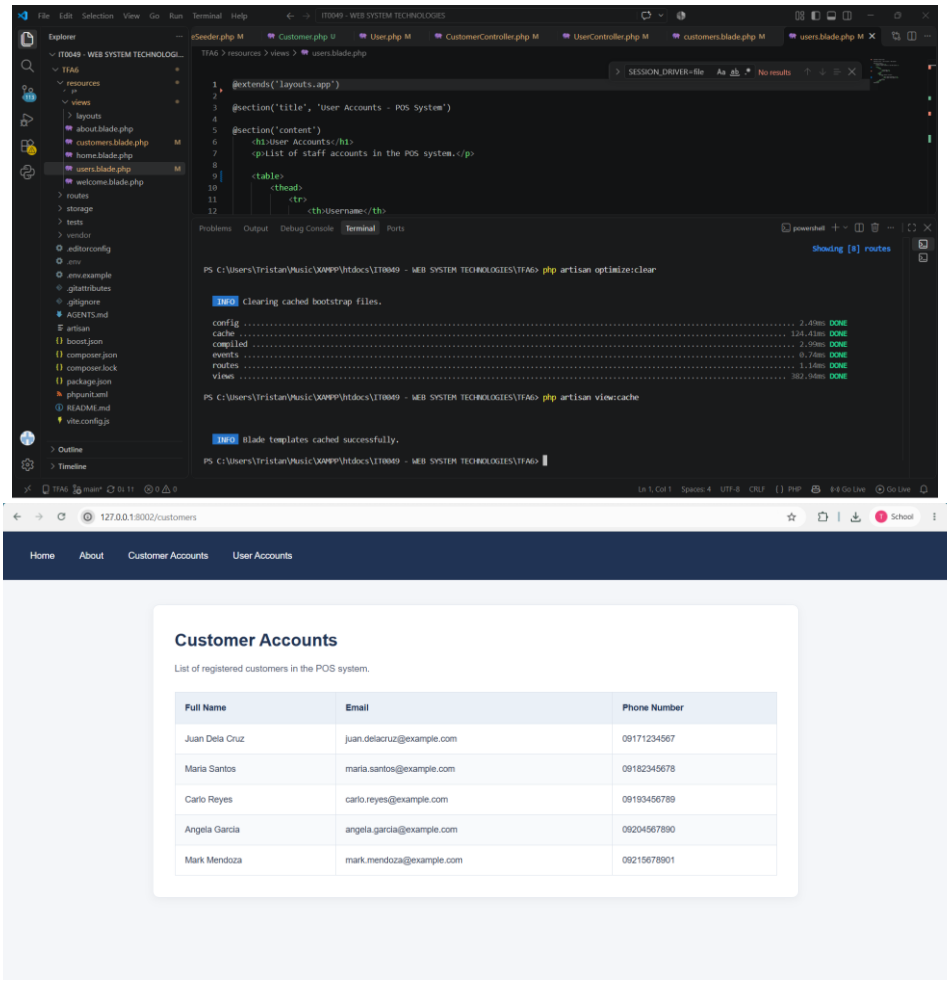
4. Controller Updates

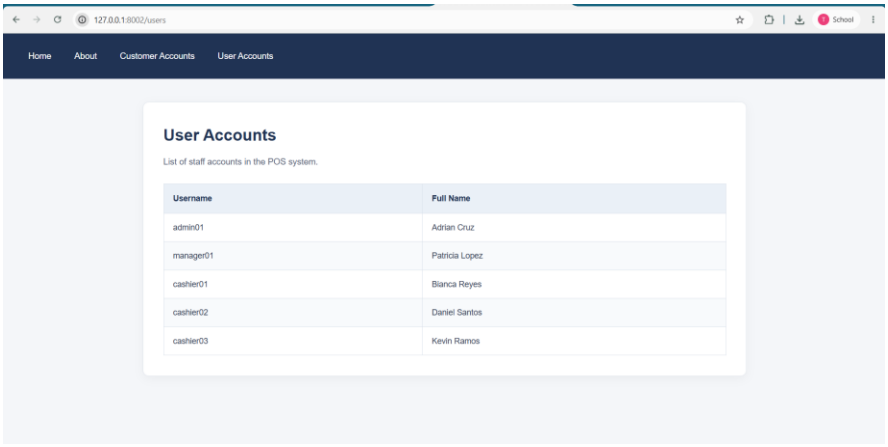
Updated CustomerController and UserController to retrieve records through Eloquent ORM instead of static PHP arrays. Both controllers pass the retrieved records to their respective Blade views. Syntax checks and Laravel route verification completed successfully.



5. Blade Views and Verification

Updated the Customer Accounts and User Accounts Blade views to display records retrieved through Eloquent ORM using @foreach loops. Both pages successfully displayed five records from the SQLite database. The existing navigation and CSS design were preserved, and Blade template caching completed successfully.




127.0.0.1:8002/users
School
Home
About
Customer Accounts
User Accounts
User Accounts
List of staff accounts in the POS system.
Username
Full Name
admin01
Adrian Cruz
manager01
Patricia Lopez
cashier01
Bianca Reyes
cashier02
Daniel Santos
cashier03
Kevin Ramos